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Lightweight authentication for IoT devices (LAID) in sustainable smart cities

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As smart cities evolve to meet sustainability goals; the proliferation of Internet of Things (IoT) devices presents both opportunities and challenges. Due to constrained nature of IoT, ensuring security in IoT enabled network is a challenge. However, growing urban population and exponential rise in IoT enabled devices make it challenging to implement and ensure security mechanisms. Several attacks such as device compromise, hijacking, tampering, man in the middle attack, replay attack, insider attack may lead to security breach. The reason behind these kinds of attacks is that an adversary has successfully obtained credentials for authentication. Several algorithms in smart cities domain have been proposed but most of them are computationally intensive schemes and some algorithms do not have password update feature. In this paper, we propose a 4-phase lightweight authentication algorithm for IoT devices in sustainable smart cities (SSC) utilizing elliptic curve cryptography (ECC) with password update phase and improved computational and communication cost. We perform a formal security analysis of the proposed algorithm using AVISPA tool and informal semantic analysis to demonstrate that our algorithm is secure against different types of attacks including MITM attack, insider attack, replay attack smart device capture attack and others. The proposed algorithm shows remarkable improvement having 1.248 ms computational time and 1036 bits communication cost. The performance analysis shows that the proposed algorithm is lightweight and shows 22.19%, 28.05% and 22.19% improvement in computation, communication and energy costs respectively as compared to the closest counterpart in existing state-of-the-art approaches.

Keywords Authentication, IoT security, Sustainable smart cities, Smart cities, Sustainability, Energy efficiency

The world is rapidly urbanizing and by 2050, the United Nations projected that nearly 70% of the global population will call start living in urban cities. Expectedly, by 2050, around 70% people may live in urban areas out of total expected population of 9.7 billion[1].

To address the growing demands on resources and infrastructure, cities are embracing a new paradigm called as smart cities. Smart cities leverage technology to collect and analyze data from a network of sensors and connected devices. A McKinsey Global Institute study suggests smart cities could generate up to $11.1 trillion annually by 2025 [2]. Internet of Things (IoT) devices are responsible to connect and collect data for providing efficient services. It is a network of physical connected devices that enable these devices to connect and exchange data. IoT devices are constrained devices having limited memory space such as RAM (8-32KB), ROM (48-512KB) and CPU with low computational power such as 8–96 MHz. IoT devices transmit data on low bandwidth connection with data rate ranging from 16 to 250 kbps[3]. Smart cities use IoT devices to collect and process data, such as smart meters, connected sensors and smart bulbs. These data are used by the cities to enhance public utilities, infrastructure, improve resource utilization and other services. With the help of IoT devices, smart cities optimize resources usage, enhance public services, improve safety, increase productivity and automation etc[4]. Smart cities are not only about technology but also about using data to create a more sustainable, efficient, and livable urban environments. As cities around the world struggle with urbanization, smart solutions offer a promising path forward.

It is evident that, smart city paradigms viz. smart environment and smart mobility are connected with environmental aspect of SSC, smart economic, smart governance and smart agriculture is connected to economic aspect of SSC and lastly, smart education, smart healthcare and smart living is connected to social aspect of SSC. However, recent developments in sustainable smart cities face several issues such as device-to-

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device communication between IoT devices, interoperability, user authentication etc. Several attacks such as device compromise, hijacking, tampering, man in the middle attack, replay attack and insider attack may lead to breach in authentication of IoT devices. Therefore, secure authentication is of utmost importance especially in IoT network due to the significance of data and utility provided by the devices. Usually, the IoT devices have constrained nature due to the characteristics of its operational functionality.

The remaining sections of this paper are structured as follows: Related work discusses related works, research contributions and motivation. Proposed algorithm presents the proposed algorithm including preliminaries, system model and threat model. Security analysis discusses the automated formal security analysis and informal analysis of our algorithm. Results and performance analysi presents functional analysis and compares the security features, computational cost, communication cost and energy consumption of proposed algorithm with other existing algorithms. Lastly, Conclusion and future work presents the conclusion and future research work.

Related work

In this section, we discuss state-of-the-art literature related to various authentication scheme among IoT devices in different smart city domains. Within the last decade, enormous researches have been conducted related to smart cities and/or sustainable smart cities including security and privacy in several domains including IoT, IIoT, Healthcare, Smart Cities, Smart Grid, IoV, VANET etc.

This section emphasizes on network domains of IoT devices and performance of the different authentication schemes. Different network domains include IoT (such as IIoT, WSN etc.), Smart Cities (such as Smart Grid, Smart Home etc.), IoV (such as IoD, VANET etc.) and Healthcare (WBAN). We have focused our study on existing literature mainly on authentication among IoT devices in different domains of smart city. The comprehensive analysis of existing authentication schemes across various IoT domains is presented in Table 1 consisting of 'Proposed method', 'Functions Used', 'Computational Costs (ms), 'Communication Cost (bits)', 'Research Findings/ Shortcomings', 'Security Analysis' and 'Threat Model'.

There are numerous published papers reviewing different authentication schemes and challenges faced in domain of IoT and smart cities which are mentioned in references[5–11]. In[12]the authors presented physical unclonable function and biometric based authentication scheme. In[13]the authors utilized biometric and fuzzy functions to develop an authentication scheme for smart cities. The authors of[14] have given authentication scheme among IoT devices in smart grid network. However, the scheme is based on PUF and is therefore prone to physical attacks, is computationally intensive and lacks a password update phase. The authors of[15] have presented an authentication scheme among devices in IoT network. However, the scheme is based on Fuzzy Extractor function and is computationally intensive with no password update phase.

In IoT networks, studies[15–25] have proposed multi-phase authentication schemes with varying computational costs ranging from 0.015824 ms to 227 ms, but most suffer from the absence of password update mechanisms and vulnerability to insider attacks. Several other schemes have been proposed in[26–28] but most of them have computationally intensive schemes and lack password update functionality.

Smart home authentication schemes presented in the related work[26,29–31]offers relatively lower communication costs of 464 bits to 2848 bits, however, they lack password update phase and key management challenges are prevalent for large-scale device deployments.

Smart grid authentication schemes presented in the related work[14,32–38] have different computational costs from 0.01105 ms to 102.358 ms, but these schemes lack password update functionality.

IIoT authentication approaches presented in the related work[28,39,40]have high computational costs and storage overhead concerns. Schemes presented in the related work for smart cities[41–43] have communication costs ranging from 1152 bits to 2560 bits, but they also do not have password update phase.

Schemes presented in the related work for vehicular network[44,45] and healthcare[46–48] also show high computational costs of 58.8955 ms and 7.5 ms respectively, making them unsuitable for resource-constrained environments.

In mobile authentication scheme presented in related work[49] are incompatible with constrained devices despite moderate computational requirements.

In wireless sensor network, authentication scheme presented in the related work[50] shows high communication costs of 3500 bits due to biometric-based authentication.

Recently, the schemes presented for authentication of IoT devices are presented in[51–53]. In[51]fuzzy verifier is used for two factor authentication using ECC having computational cost of 8.9 ms and communication cost of 2496 bits. The scheme has computational overhead and synchronization issues due to the presence of fuzzy verifier. The authors of[52,53] presented authentication scheme based on blockchain involving decentralized registration of devices.

The identified research gaps from the extensive literature review include the absence of password update phase, high computational and communication costs unsuitable for constrained IoT devices, vulnerability to various attacks including password guessing and insider attacks, certificate management complexities, and scalability issues. Many schemes also rely on hardware-based secure elements that introduce integration overhead, while others use PUF-based approaches that are susceptible to physical attacks. Key management issues, certificate problems, key escrow challenges also bring limitations in these existing algorithms. These gaps collectively highlight the need for lightweight and secure authentication solutions that can address the unique constraints of IoT environments while maintaining security properties. Therefore, a lightweight and secure authentication algorithm for IoT devices remains an open research problem.

Motivation

Evidently, IoT devices plays a vital role in the development and realization of sustainable smart cities. The integration of IoT technologies enables cities to becoming more resilient, effective and environmentally friendly

Reference	Domain/ Network Type	Proposed method	Functions Used	Computational Costs (ms)	Communication Cost (bits)	Research Findings/ Shortcomings	Security Analysis	Threat Model
14	Smart Grid	PUF based 3 phase scheme for authentication	HASH, XOR, PUF	1.09	1536	Password update phase not available, PUF based, computationally intensive, vulnerable to physical attack	Informal & Formal using SVO	CK
15	IoT	2 factor 5 phase scheme for authentication	ECC, HASH, XOR	15.71	2432	Complex and computationally intensive	Informal & Formal using BAN Logic	DY
26	Smart Home	2 phase scheme for authentication	ECC, HASH, XOR, CRT	312	525	Password update phase not available, Key management issue for large number of devices due to CRT	Informal	-
27	IoV	PUF based scheme for authentication	Service Based Categorization	431.999	-	PUF based, computationally intensive, high complexity	-	-
16	IoT	3 phase scheme for authentication	ECC, HASH, XOR	8.935	1280	Password update phase not available	Informal & Formal using BAN Logic	DY
17	IoT	4 phase scheme for authentication	ECC, HASH, XOR, Elgamal	19	1024	Password update phase not available, computationally intensive	Informal	DY
28	IIoT	2 phase scheme for mutual authentication	ECC, HASH, XOR, PK E/D, Symmetric E/D	49.469	NA	Password update phase not available, computationally intensive, hardware based secure element integration overhead	Informal & Formal using ROM	DY
18	IoT	3 phase authentication and key agreement scheme	HASH, XOR	20.148	1152	Password update phase not available, less secure	Informal & Formal using ROM	DY
19	IoT	3 phase scheme for authentication	ECC, HASH, XOR	0.015824	1760	Password update phase not available, vulnerable to anonymity, password guessing and insider attacks	Informal & Formal using AVISPA	DY
20	IoT	3 phase scheme for authentication	ECC, HASH, XOR	0.017824	1760	Password update phase not available, vulnerable to offline password guessing and insider attacks.	Informal & Formal using AVISPA	DY
39	IIoT	4 phase scheme for authentication	ECC, HASH, XOR, Fuzzy function	387.8	4416	Storage overhead, computationally intensive	Informal & Formal using ROM	DY
21	IoT	4 phase scheme for authentication	HASH, XOR	21.91	3440	Less secure, computationally intensive, mathematical functions XOR and HASH are used	Informal & Formal using ROM	DY
40	IIoT	PUF based 4 phase scheme for authentication	HASH, XOR	0.39	1536	Password update phase not available, no forward and backward secrecy	Informal	-
22	IoT	4 phase scheme for authentication	ECC, HASH, XOR	108	2048	Computationally intensive	Informal & Formal using ROM	DY
23	IoT	PUF based 4 phase scheme for authentication	ECC, HASH, XOR	11.296	-	PUF based, computationally intensive	Informal & Formal using ROM	DY
24	IoT	3 phase scheme for authentication	HASH, XOR, ECC, Symmetric E/D, Fuzzy Function	227	1920	Password update phase not available	Informal & Formal using ROR	-
41	Smart Cities	3 phase scheme for authentication	ECC, HASH, XOR	0.13744	1152	Password update phase not available, computational overhead in each session	Informal	-
42	Smart Cities	4 phase scheme for authentication	HASH, XOR	6.93	2560	Less secure, computationally intensive	Informal & Formal using BAN Logic	DY
54	Smart Cities	6 phase authentication and key agreement scheme	HASH, XOR, PK E/D	0.53	-	Lightweight	Informal & Formal using BAN Logic	-
43	Smart Cities	4 phase authentication and key agreement scheme	HASH, XOR, ECC, Symmetric E/D, PK E/D	4.322	2336	Password update phase not available, computationally intensive	Informal & Formal using AVISPA	-
29	Smart Home	4 phase scheme for authentication	ECC, HASH, XOR	68.425	464	Mutual authentication among the three entities	Informal & Formal using BAN Logic	DY
30	Smart Home	Fuzzy extractor based 5 phase scheme for authentication	HASH, XOR	0.0524	1216	Password update phase not available, less secure	Informal & Formal using AVISPA	DY
31	Smart Home	6 phase authentication and key agreement scheme	ECC, HASH, XOR,	0.55	2848	Communication overhead, computationally intensive	Informal & Formal using ROR	-
44	IoD	Certification based 3 phase scheme for authentication	ECC, HASH, XOR	1.604	2408	Certification based scheme	Informal & Formal using Proverif	DY

Continued

Reference	Domain/ Network Type	Proposed method	Functions Used	Computational Costs (ms)	Communication Cost (bits)	Research Findings/ Shortcomings	Security Analysis	Threat Model
45	VANET	6 phase authentication and key agreement scheme	ECC, HASH, XOR	58.8955	1152	Complex and computationally intensive	Informal & Formal using ROM	DY
46	WBAN	3 phase scheme for authentication	ECC, HASH, XOR	2.88	1440	High complexity and scalability issues due to heterogenous authentication mechanism	Informal & Formal using ROM	DY
47	Healthcare	4 phase scheme for authentication & key establishment	HASH, XOR	0.0448	1152	Password update phase not available, less secure	Informal & Formal using BAN Logic/ AVISPA	DY
48	Healthcare	5 phase authentication scheme	HASH, XOR, ECC, PK E/D, Fuzzy function, PUF, blockchain	7.5	2911	Complex and computationally intensive	Informal & Formal using ROR/ AVISPA	-
32	Smart Grid	3 phase scheme for authentication	ECC, HASH, XOR	0.026	1440	Password update phase not available, Self-certification is trust challenging	Informal & Formal using ROM	CK
33	Smart Grid	3 phase scheme for authentication	ECC, HASH, XOR	0.012726	544	Password update phase not available, certificate management and key escrow problem	Informal	CK
34	Smart Grid	3 phase scheme for authentication	HASH, XOR	0.01105	960	Only mathematical functions XOR and HASH used	Informal & Formal using ROR/ AVISPA	DY
35	Smart Grid	3 phase scheme for authentication	ECC, HASH, XOR, Symmetric E/D	5.319	1056	Password update phase not available, Limited security proof	Informal & Formal using ROM/ AVISPA	DY
36	Smart Grid	2 phase scheme for authentication	HASH, XOR, Symmetric E/D	0.39	1088	Password update phase not available, Key management issue for large number of devices due to Symmetric Key E/D	Informal	-
37	Smart Grid	4 phase scheme for authentication	ECC, HASH, XOR	102.358	962	computationally intensive	Informal & Formal using ROM	DY
38	Smart Grid	5 phase key management scheme	HASH, XOR, PK E/D	3.1	1024	Password update phase not available, less secure	Informal & Formal using ROM	-
25	IoT	Protocol base 3 phase scheme for authentication	ECC, HASH, XOR, Symmetric E/D	25.06	1600	BAN logic is not suitable for verifying key exchange protocols	Informal & Formal using SVO/ AVISPA	DY
49	Mobile	2 factor 4 phase scheme for authentication	ECC, HASH, XOR	1.818	1376	Not suitable for constrained devices	Informal & Formal using ROM	DY
50	WSN	6 phase authentication and key agreement scheme	ECC, HASH, XOR	54.2	3500	High complexity and computationally intensive due to biometric based authentication	Informal & Formal using BAN Logic	DY
51	IoT	2 Factor authentication using ECC and Fuzzy verifier	HASH, XOR, ECC, PK E/D, Fuzzy function	8.95	2496	Computationally intensive and synchronization issues due to the presence of fuzzy verifier	Informal & Formal using ROM	DY
52	IoT	Blockchain based identity authentication scheme	ECC, HASH, XOR, blockchain	13.34	1760	Password update phase not available, decentralized registration of devices	Informal	-
53	IoT	Framework for mutual authentication using private Ethereum	ECC, HASH, XOR, blockchain	1.89	2080	Key management issues and vulnerable to may lead to device capture attack, stolen-verifier attack	Informal & Formal using BAN Logic	-

Table 1. Related work.

while enhancing the quality of life. By leveraging the data and insights provided by IoT technologies, cities can make informed decisions, optimize resources and implement strategies that address environmental challenges, promote sustainability and enhance the overall quality of life for their residents. However, recent developments in sustainable smart cities face several issues in areas like device-to-device communication between IoT devices, interoperability, user authentication etc. Prevalent issues in IoT devices operating in different paradigms of smart cities include weak authentication mechanism, insecure communication, privacy concerns, insider attacks, physical tampering, man in the middle attack, replay attack, insider attack etc. The reason behind these kinds of attacks is that an adversary has successfully obtained credentials for authentication. Therefore, a secure and lightweight authentication with lower computation and communication cost is of utmost importance in IoT ecosystem due to the significance of data and utility provided by the devices.

Research contributions

In view of the research problem, this paper makes the following contributions:

- We utilize ECC to propose a lightweight authentication algorithm for securing IoT devices in sustainable smart cities. The proposed algorithm generates nonce to initiate the computations and verification of exchanged messages and utilizes the security strength of Elliptic Curve Discrete Logarithm Problem (ECDLP). We have added a password update phase to change the password of the IoT devices by the user.
- We focus on minimum message exchanges considering the constrained nature of IoT devices. Also, during authentication, session may be terminated if any verification is not true.
- We elaborate formal automated security analysis utilizing AVISPA tool and using OFMC and CL-AtSe backend models[55].
- We perform extensive semantic security analysis to demonstrate that our algorithm supports device anonymity, forward and backward secrecy, key freshness, mutual authentication, non-traceability and is resistant towards cookie theft attack, impersonation attack, insider attack, MITM attack, smart device capture attack and other attacks is demonstrated.
- Comparative performance parameter like computational cost, communication cost and energy consumption are calculated and evaluated to show that our algorithm incurs lower computation, communication and energy costs.

Proposed algorithm

This section provides the detailed concept of the proposed approach including preliminaries, system model, threat model etc. and our proposed algorithm.

Preliminaries

This section mainly introduces the preliminaries for improving the readability of the research paper.

Elliptic curve cryptography

ECC was first introduced by Koblitz[56] and Miller[57]. ECC is a type of public-key cryptography based on the algebraic structure of elliptic curves over finite fields. Further reading on ECC can be found in[58].

Elliptic curve discrete logarithm problem (ECDLP)

Mainly, the strength of ECC security depends on the computational difficulty of solving ECDLP. ECC comprises of smaller key sizes compared to other public-key cryptosystems resulting in faster computations and lower power consumption. Hence, ECC incorporates increased security strength for the same key size compared to other cryptosystems. ECC algorithms and parameters are standardized by organizations like NIST and ANSI[58].

Definition 1 Given two points P and Q over $E_p\,(a,\ b)$, the elliptic curve discrete logarithm problem (ECDLP) is to find an integer $s \epsilon\ F_p$ such that $Q = s.P$.

The elliptic curve discrete logarithm problem can be stated as follows. For any prime p and an elliptic curve $E_p\,(a,\ b),\ Q = s.P$, Then, the ECDLP is to determine s for given P and Q. It is relatively easy to calculate Q for given s and P, but it is computationally infeasible to determine s for given Q and P.

Pollard-Rho algorithm solves ECDLP in exponential time with time complexity of $\sqrt{\prod\ {}^{*}\mathrm{n}\ /2}$[59,60].

Mathematical functions

Hash function can be defined as $h : \{0,1\}\ \rightarrow\ \{0,1\}^n$ as defined in[61]. Hash function takes input $x \epsilon\ \{0,1\}$ and gives output as $h\,(x)\ \epsilon\ \{0,1\}^n$ of a fixed length n. The properties of hash function are given below:

- $h\,(x)$ can be applied on any and every part of data.
- $h\,(x)$ gives fixed length output.
- $h\,(x)$ is called one way function because it is computationally infeasible to compute input x, if we have output and $h\,(x)$
- $h\,(x)$ is called weak-collision function because it is computationally infeasible to compute any other input x_2, such that $x_2 \neq\ x_1$ such that for $h\,(x_1) = h\,(x_2)$.
- $h\,(x)$ is called strong-collision resistant function because it is computationally infeasible to determine two different input messages $x_1,\ x_2$ that produces the same hash value such that $h\,(x_1) = h\,(x_2)$

XOR (eXclusive OR) function is basic function used in cryptographic algorithm for encryption and decryption. XOR function generates single binary output from two binary inputs. Output bit of XOR function will be 1 for different input bits whereas it will be 0 for same input bits.

AVISPA

We have used AVISPA (Automated Validation of Internet Security Protocols and Applications) for validation and formal analysis of our algorithm[62]. AVISPA is a tool for the automated validation of security protocols and applications. SPAN (Security Protocol Animator) is a component of the AVISPA tool suite. SPAN is a graphical user interface (GUI) tool part of the AVISPA tool suite. SPAN provides a graphical representation of the protocol, making it easier to visualize and understand the flow of messages and actions. The graphical interface displays the roles, message exchanges, and security goals of the protocol. SPAN can animate the execution of the specified protocol, allowing users to step through the protocol and observe the message exchanges and state transitions. It provides a user-friendly environment for specifying security protocols and running validation tests. SPAN allows users to specify security protocols using the High-Level Protocol Specification Language (HLPSL).

HLPSL is a role-based language that describes the behavior of protocol participants and their interactions. HLPSL is translated to intermediate format (IF). Users can initiate the validation process from within SPAN, which translates the HLPSL specification into an intermediate format and passes it to the backend analysis tools. On execution, SPAN integrates with the other components of AVISPA, such as the On-the-Fly Model-Checker (OFMC) and the Constraint-Logic-based Attack Searcher (CL-AtSe)[63] to validate security checks by converting translated IF into output format (OF). The analysis results including potential attacks or security flaws are displayed in SPAN's output window.

System model

Before introducing the proposed lightweight authentication algorithm for securing IoT devices in sustainable smart cities, we introduce exemplary smart home paradigm of sustainable smart cities as shown in Fig. 1. Smart home environment has multiple smart appliances and/or devices having IoT/wireless sensor capabilities. The smart home system model generally comprises the following entities:

- Home Server (HS): HS is a trusted entity responsible for authentication of IoT enabled devices.
- Devices: Devices are smart home appliances enabled with IoT/sensing capabilities for transmitting dynamically sensed information through the gateway to users.
- Users: Users are humans who use these devices in smart home environment.
- Gateway: Gateway is an intermediary device for connecting IoT devices and users to establish connection and communicate information among the devices and users.

Adversarial threat model

We have used Dolev–Yao (DY) threat model[64] as widely recognized attacker model in security analysis. In DY threat model, any two entities communicating on an open and insecure channel cannot be treated as trusted entity and information is transmitted on open and insecure channel. DY threat model is used to find logical errors in the security protocol. In DY model, the communicating entities are not trusted and transmission channel is not secure. DY threat model creates a formal verification method based on symbolic analysis. It has been widely accepted and used to find vulnerabilities in the security protocol.

In DY threat model, the adversary $\mathscr{A}$ has enhanced attacker capabilities, the adversary's capabilities can be listed as follows:

- $\mathscr{A}$ exercises complete dominion over the network, enabling them to eavesdrop, obstruct, and intercept any message traversing the communication channels. Furthermore, $\mathscr{A}$ has meticulously recorded all previously exchanged messages between honest entities.
- $\mathscr{A}$ possesses the capability to send and resend messages at their discretion, as well as combine and decompose messages in a manner that suits their objectives.
- $\mathscr{A}$ can perform any encryption operation specified within the protocol's parameters and decrypt encrypted messages, provided they possess the requisite decryption key.

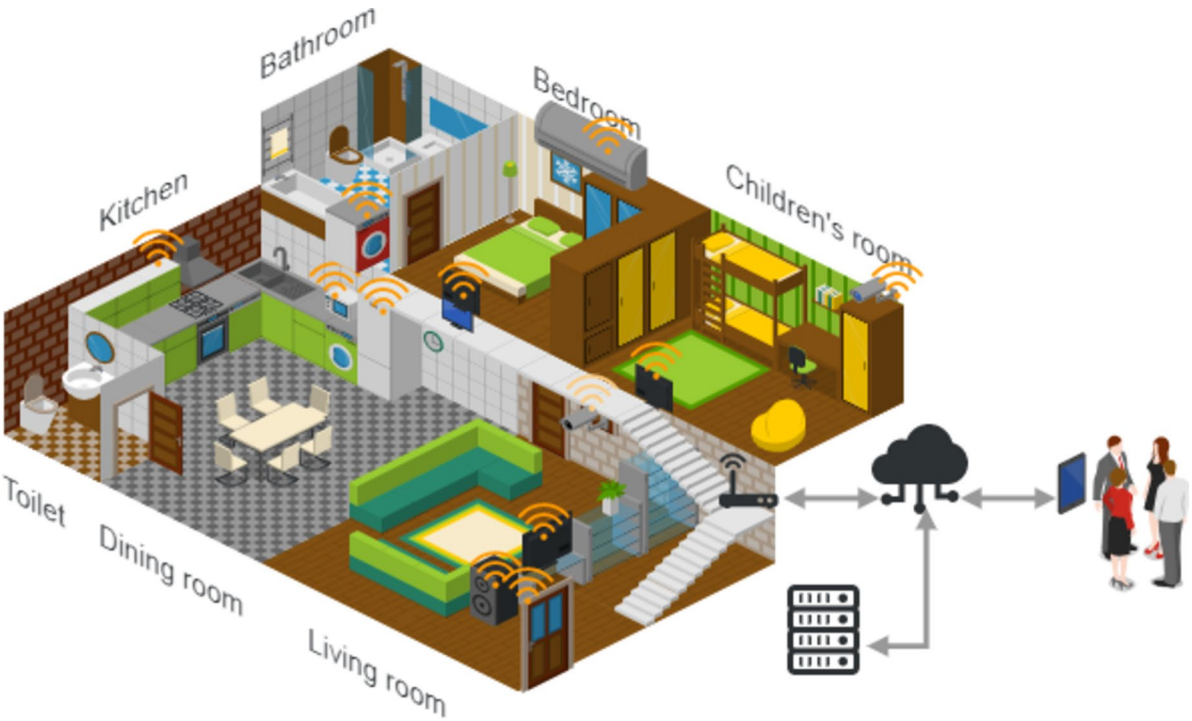


Fig. 1. Exemplary system model.

- $\mathscr{A}$ may be a legitimately registered device within the system, granting them access to all security parameters and associated privileges.
- $\mathscr{A}$ could potentially be a legitimate yet malicious device, allowing them to obtain sensitive information from a limited number of sensor nodes within the network.

Notations and symbols

The section defines important notations and symbols in Table 2 used in the research paper for better readability and understanding.

Lightweight authentication for IoT devices (LAID) algorithm – proposed approach

In this section, a Lightweight Authentication for IoT Devices (LAID) algorithm is proposed, applicable to all paradigms of Sustainable Smart Cities. LAID is a 4-phase algorithm namely initialization phase, device setup phase, authentication phase and password-change phase. The adversarial threat model is also described with the attacker capabilities and open channel network comprising DY model. The security strength of LAID is based on the difficulty of ECDLP as defined in Definition 1.

The different phases of LAID algorithm are described below. After the authentication phase, both Home Server *(HS)* and Device (D_i), establish a secret key K_{LAID}. Different notations and symbols are summarized in the Table 2.

Algorithm

LAID algorithm consists of 4 phases, sequence diagram is shown in Fig. 2 and a flow diagram is shown in Fig. 3. In this section, a detailed description of the algorithm is presented considering the exemplary system model of typical smart home as shown in Fig. 1.

Initialization Phase: In this phase, the Home Server *HS* selects and broadcasts the global parameters *(p, E, S, l)* required for elliptic curve operations. All the devices who are willing to establish connection with the Home Server receives the broadcasted global parameters.

Device Setup Phase: In this phase, the Home Server *HS* computes W and HSD_{MSG}, using its symmetric key and sends W and HSD_{MSG} to the device D_i willing to establish communication.

Authentication Phase: In this phase, a mutually authenticated key for the session is established between the Home Server *HS* and the device D_i. This phase is initiated from the device D_i, which is logged in by using a password P_D.

Step 1: The device D_i selects a pseudorandom number $nonce_D$ and computes S_D and sends HSD_{MSG} and S_D to the *HS*.

Step 2: On receiving HSD_{MSG} and S_D, the *HS* verifies the received HSD_{MSG} after decrypting with HSD_{SYM_KEY}. This is required to verify the identity of the device D_i who is willing to establish the connection. If the verification is false, the session will be terminated. Otherwise, the HS computes S_{HS}, K_{LAID} and N_{HS}. Then, the *HS* sends S_{HS} and N_{HS} to the device D_i.

Step 3: The device D_i computes W' and K_{LAID} and verifies the received N_{HS}. If the verification is false, the session will be terminated. Otherwise, the device D_i computes N_D and sends to the HS.

Notation/Symbol	Meaning	Notation/Symbol	Meaning
||	Concatenation Operation	XOR, $\oplus$	Mathematical XOR operation
==	Equivalence Relation	$\mathscr{A}$	Adversary
p	Prime number in finite field F_p	E	Elliptic Curve $E(F_p)$
S	Subset of points on E	l	Order for generating subset S
HS	Home Server	D_i	Smart Device
i_D	Unique Identifier associated to each device D_i	P_D	Password set by the HS
h(.)	Hash Function	E	Symmetric Encryption Operation
HSD_{MSG}	Encrypted message computed by the HS	HSD_{SYM_KEY}	Secret Key of HS
nonce	Pseudorandom number	$nonce_D$	Pseudorandom number selected by device
$nonce_{HS}$	Pseudorandom number selected by the HS	K_{LAID}	Mutually authenticated key for the session between the HS and the device
	Internet		Wifi Enabled Smart Device
	User		Home Gateway
	Connection Link		Home Server

Table 2. Notations and symbols.

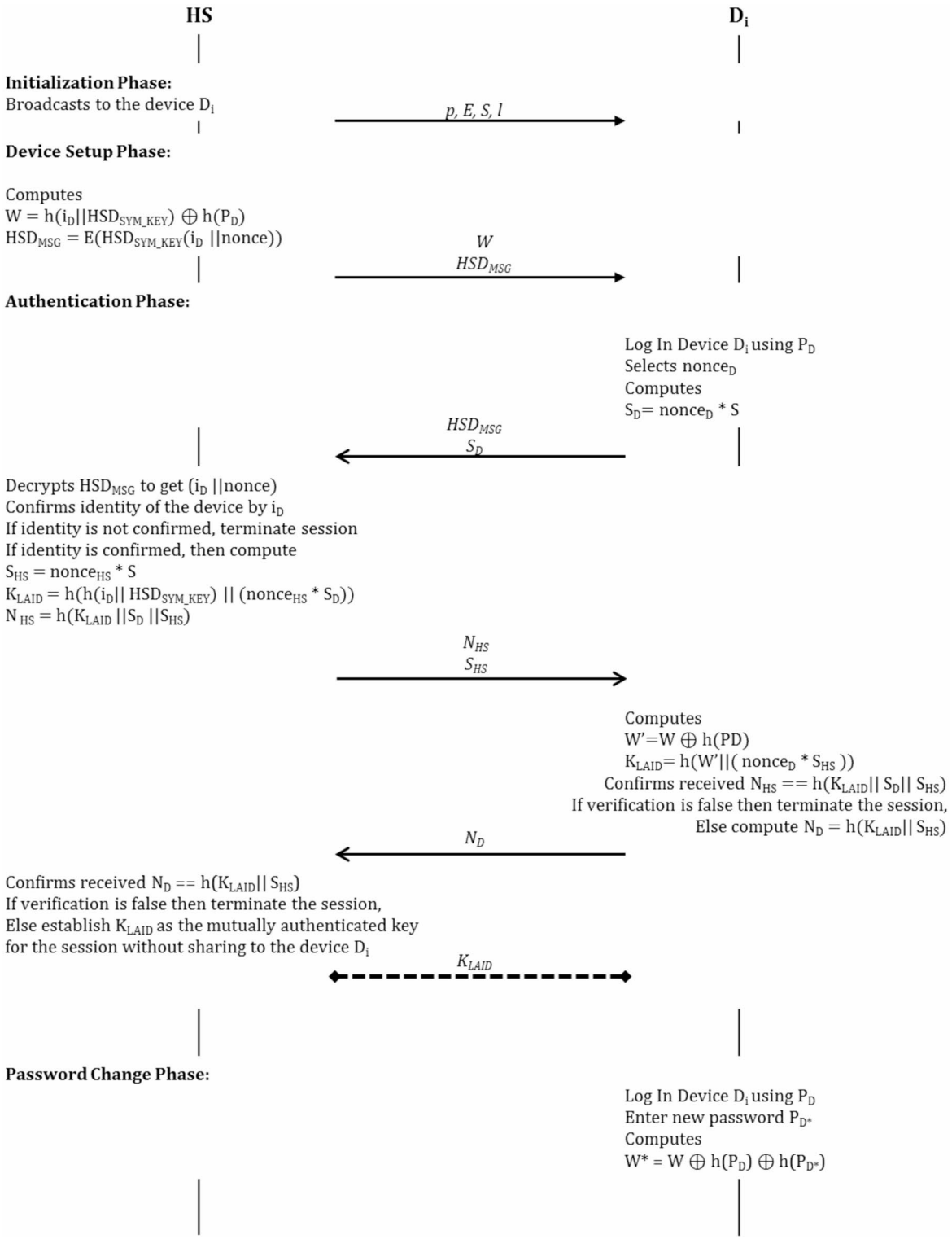


Fig. 2. Sequence diagram of LAID algorithm.

Step 4: The *HS* verifies the received N_D and if the verification is false, it will terminate the session. Otherwise, K_{LAID} is established as mutually authenticated key for the session between the *HS* and the device D_i.

Hence, the mutually authenticated key is established after three message exchanges incurring lower communication cost also.

Password-Change Phase: In this phase, the device D_i, is logged in by using the existing password P_D and enter the new password P_D*, then, the device D_i computes W* and assign it to W.

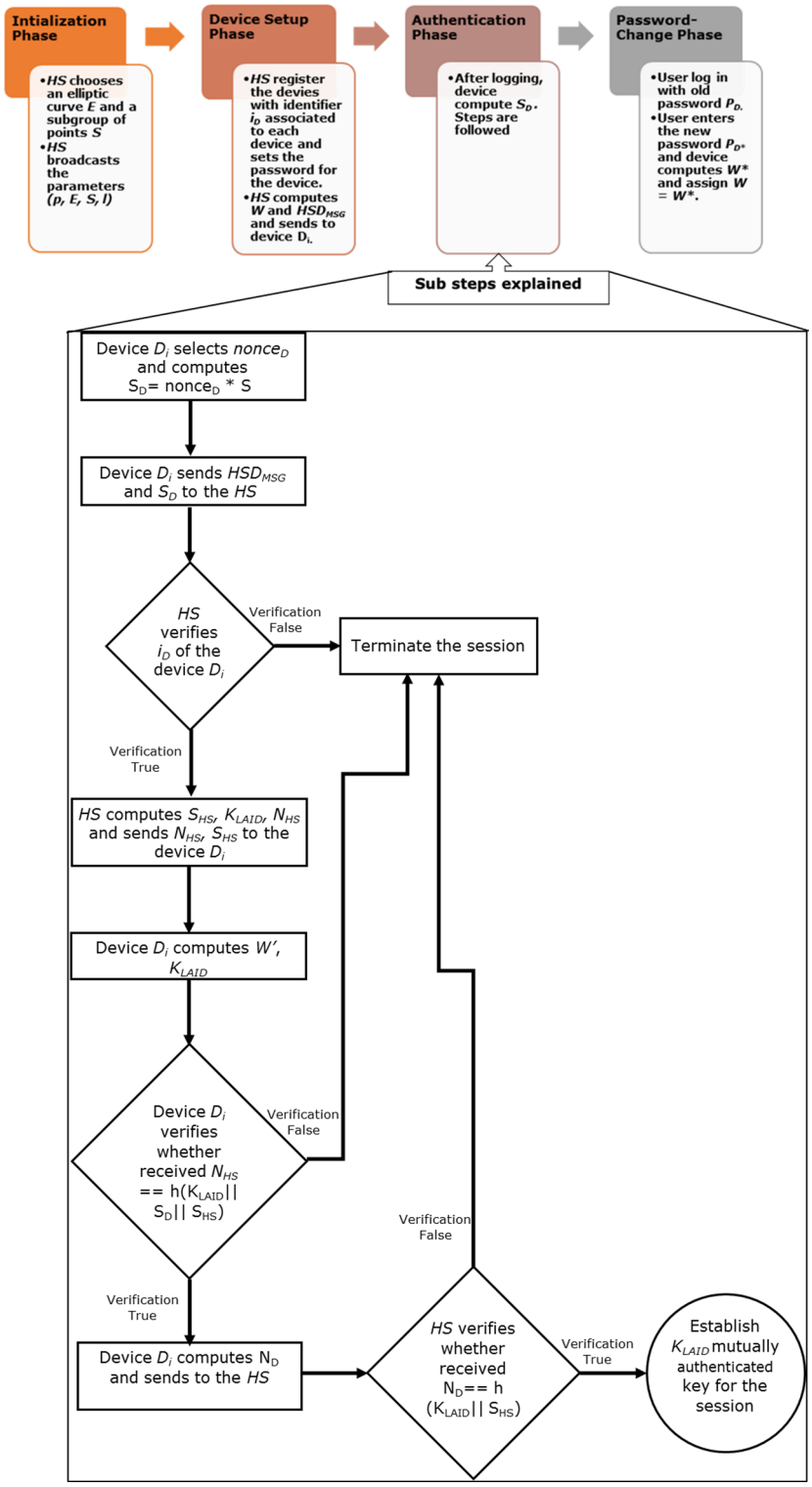


Fig. 3. Flow diagram of LAID algorithm.

Mathematical validation

Since, the device D_i computes K_{LAID} = h(h(i_D|| HSD_{SYM_KEY})||($nonce_{HS} \times S_D$)) and the *HS* computes K_{LAID} = h(W'||($nonce_D$ x S_{HS})) where W' = W ⊕ h(P_D) and W = h(i_D|| HSD_{SYM_KEY}) ⊕ h(P_D), resulting into W' = h(i_D|| HSD_{SYM_KEY}). Also, $nonce_{HS}$ * S_D = $nonce_{HS}$ * $nonce_D$ * S = $nonce_D$ * S_{HS}. Thus, established K_{LAID} is same and correct as computed individually by the *HS* and the device D_i without transmitting K_{LAID} on the open channel.

Security analysis

In this section, we perform automated formal security analysis and informal security analysis of the proposed LAID algorithm.

Automated formal security analysis

AVISPA, as described in preliminaries in Preliminaries, is used as virtual environment[62,63,65] to formally verify the LAID algorithm. In AVISPA, LAID algorithm is written in high level language called HLPSL, then HLPSL program is translated by HSPSL2IF translator into intermediate format (IF). Further, security is validated by output format (OF) using the converted IF.

The role specifications of the Home Server *HS* and the device D_i in form of AVISPA code snippets are presented in Fig. 4 (a-d). The role of the Home Server *HS* and the device D_i is defined in the role section of the HLPSL code. The code snippets shown in Fig. 4 (a-d) also presents the session, environment and goal sections of the HLPSL code. There are 3 goals defined in the HLPSL code, first is to maintain the secrecy of the mutually authenticated key, second is to ensure the authentication of the device and third is to ensure the authentication of the Home Server. To evaluate security validation, OFMC and CL-AtSe backend are used and results for both backends are shown in Fig. 5 (a, b) respectively. The 'SAFE' summary is obtained by running the HLPSL code in both backends validating the successful execution of the algorithm in open channel network (DY) model.

Informal analysis

In this section, informal analysis has been performed showing that LAID algorithm can resist different attacks and supports security features as discussed below. In the propositions defined, it is proved that LAID algorithm is safe against well-known attacks.

Proposition 1 *LAID algorithm is resistant to brute force attack.*

Proof The algorithm involves multiple secret components such as $nonce_D$, $nonce_{HS}$, P_D, W, HSD_{SYM_KEY} which together contribute to the computation of K_{LAID}. An adversary $\mathscr{A}$ would face difficulty in deriving multiple components significantly increasing the complexity of deriving these components and decreasing the possibility of brute force attack. Hence, LAID algorithm is resistant to brute force attack.

Proposition 2 *LAID algorithm is resistant to cookie theft attack.*

Proof The LAID algorithm doesn't use session-specific values but randomly generated $nonce_D$ and $nonce_{HS}$ which is further used in computing K_{LAID}. Hence, cookie theft attack will not help in determining an information for future sessions. Therefore, the LAID algorithm has resistance against the cookie theft attack.

Proposition 3 *LAID algorithm preserves data confidentiality.*

Proof During authentication phase, the device D_i receives N_{HS} and compute K_{LAID} to verify received $N_{HS} == h(K_{LAID}||\ nonce_D||\ nonce_{HS})$. Similarly, the *HS* receives N_D and verifies received $N_D == h(K_{LAID}||\ nonce_{HS})$. Any adversary $\mathscr{A}$ cannot intercept the different components of N_{HS} and N_D over the insecure channel. Hence, in LAID algorithm, data confidentiality is achieved by the device D_i and the *HS* respectively.

Proposition 4 *LAID algorithm is resistant to DoS attack.*

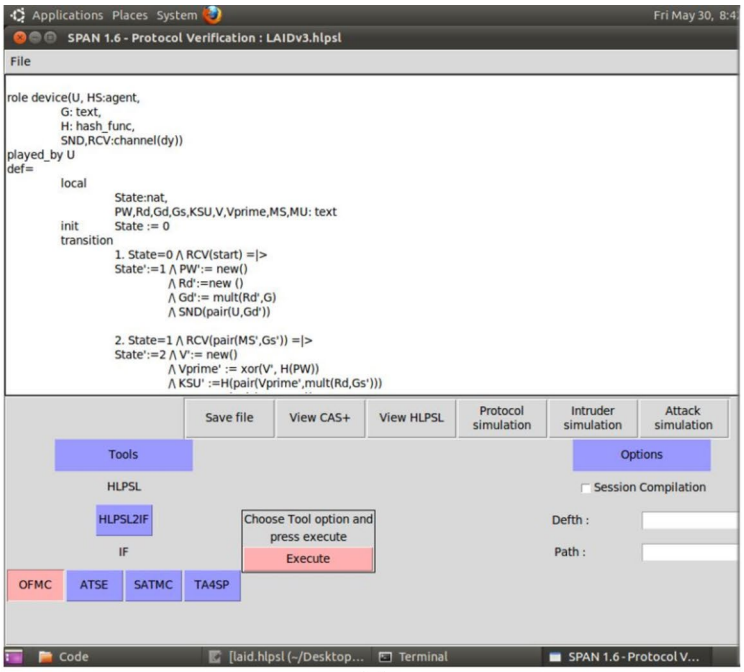
Proof In LAID algorithm, the mutually authenticated secret key is established by both the *HS* and the device D_i in a sequential manner by verifying the received messages from each other. In this process of verification and computing further components, if any verifying condition fails, then the verifying entity will realize that the other entity is illegitimate and terminate the session instantly without proceeding further. Since, communication is immediately stopped, thus preventing DoS attacks in LAID algorithm.

Proposition 5 *LAID algorithm preserves device anonymity.*

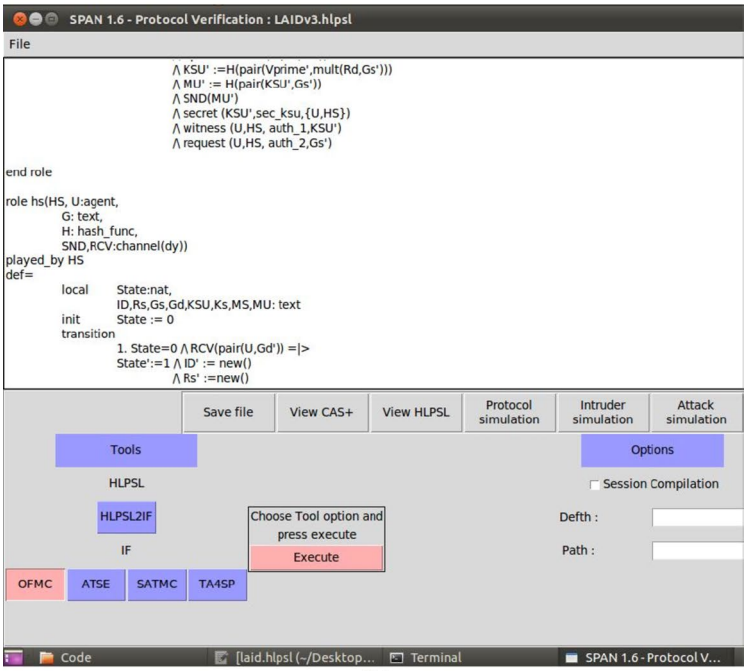
Proof In authentication phase, the device D_i selects a pseudorandom number $nonce_D$ and computes $S_D = nonce_D * S$. The device D_i sends HSD_{MSG} and S_D to the *HS*, where $HSD_{MSG} = E(HSD_{SYM_KEY}(i_D\ ||nonce))$ computed by the *HS* and sent to the device D_i in device setup phase. The *HS* already encrypts identity of the device D_i using its secret key HSD_{SYM_KEY}, pseudorandom nonces to protect identity of device and providing anonymity. Thus, any adversary $\mathscr{A}$ cannot get original identity of the device D_i and hence, LAID algorithm preserves device anonymity.

Proposition 6 *LAID algorithm preserves forward and backward secrecy.*

Proof In LAID algorithm, the device D_i and the *HS* calculate the session key as $K_{LAID} = h(W'||(\ nonce_D * S_{HS}\ ))$ computed by the device and $K_{LAID} = h(h(i_D||\ HSD_{SYM_KEY})\ ||\ (nonce_{HS} * S_D))$ computed by the HS, where, $W' = W \oplus h(P_D)$ and $W = h(i_D||HSD_{SYM_KEY}) \oplus h(P_D)$, thus $W' = h(i_D||HSD_{SYM_KEY})$. Also, $nonce_{HS} * S_D = nonce_{HS} * nonce_D * S = nonce_D * S_{HS}$, where, $nonce_{HS} * nonce_D$ are pseudorandom numbers generated by the *HS* and the device D_i. Since, it is computationally infeasible to solve ECDLP by any adversary $\mathscr{A}$, therefore, LAID algorithm preserves forward and backward secrecy.



(a)



(b)

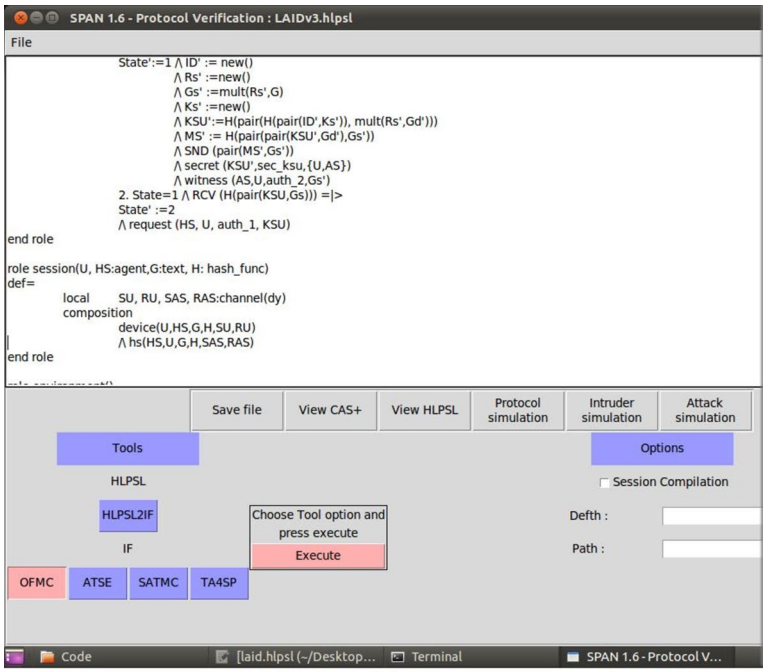
Fig. 4. (**a**) Code snippets, (**b**) code snippets, (**c**) code snippets, (**d**) code snippets.

Proposition 7 *LAID algorithm is resistant to impersonation attack.*

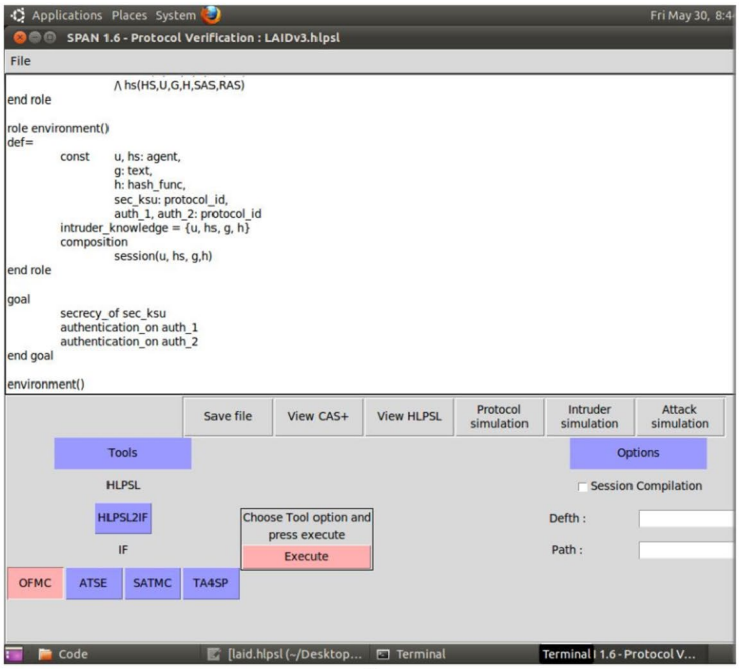
Proof Since, the private random values $nonce_{HS}$ and $nonce_D$ are not directly transmitted, and $S_D = nonce_D * S$ and $S_{HS} = nonce_{HS} * S$ makes it computationally difficult for an adversary $\mathscr{A}$ to derive $nonce_D$ and $nonce_{HS}$ needed for impersonation. Therefore, the LAID algorithm has resistance against the impersonation attack.

Proposition 8 *LAID algorithm is resistant to insider attack.*

Proof The *HS* has secret key HSD_{SYM_KEY} and any insider adversary $\mathscr{A}$ doesn't automatically gain access to the other entity's secret information. Also, the use of pseudorandom values $nonce_D$ and $nonce_{HS}$ for each session means that even if an insider compromises one session, they don't automatically gain access to other sessions.



(c)



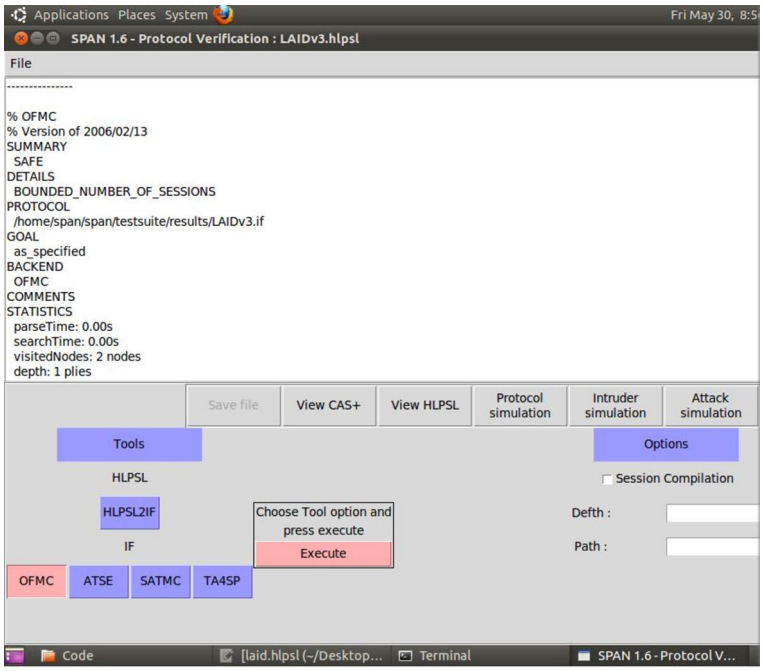
(d)

Fig. 4. (continued)

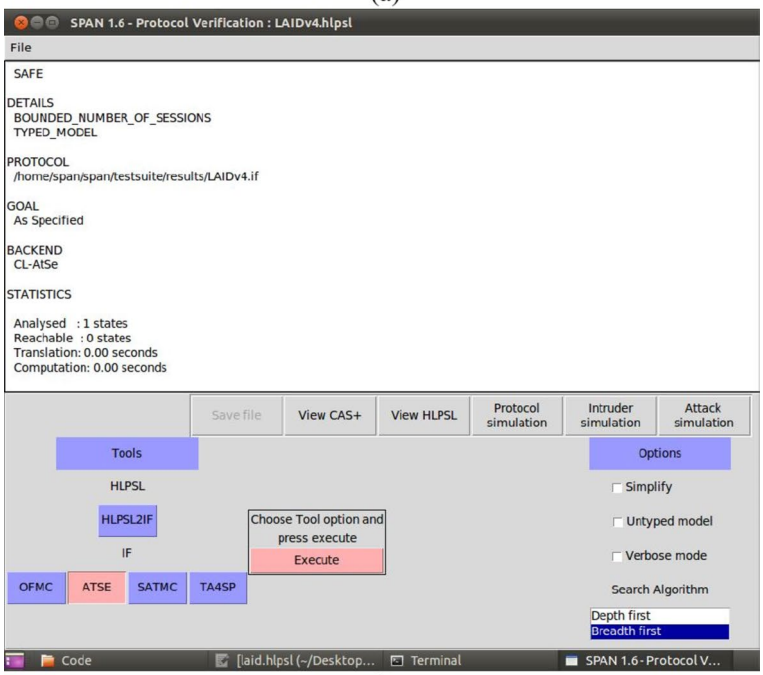
Further, the use of hash functions in computing $K_{LAID} = h(h(i_D || HSD_{SYM_KEY}) || (nonce_{HS} * S_D))$ by the *HS* and $K_{LAID} = h(W' || (nonce_D * S_{HS}))$ by the device D_i. Hence, the insider adversary $\mathscr{A}$ needs to reverse-engineer the key even if they have partial information. Therefore, the LAID algorithm has resistance against the insider attack.

Proposition 9 *LAID algorithm preserves key freshness property and prevents parallel session attack.*

Proof When any adversary $\mathscr{A}$ obtains any or all of the message HSD_{MSG}, S_D, N_{HS}, S_{HS}, and N_D to successfully login into the device D_i, however, it cannot further succeed as S_D and S_{HS} depend on pseudorandom numbers $nonce_D$ and $nonce_{HS}$ respectively. Also, for the adversary $\mathscr{A}$, it is computationally infeasible to determine the value of $nonce_D$ and $nonce_{HS}$ due to the difficulty of ECDLP. Hence, LAID algorithm preserves key freshness property and prevents parallel session attack.



(a)



(b)

Fig. 5. (**a**) Safe summary of OFMC backend. (**b**) Safe summary of CL-AtSe backend.

Proposition 10 *LAID algorithm is resistant to MITM attack.*

Proof When an adversary $\mathscr{A}$ reads the messages over the insecure channel and poses as legitimate entity, MITM attack occurs, however, K_{LAID} is derived using a hash function and it is difficult to obtain K_{LAID} even if the adversary intercepts the S_D and S_{HS}. Hence, LAID algorithm is resistant to MITM attack.

Proposition 11 *LAID algorithm preserves mutual authentication.*

Proof In authentication phase, the *HS* computes $K_{LAID} = h(h(i_D || HSD_{SYM_KEY}) || (nonce_{HS} * S_D))$ and $N_{HS} = h(K_{LAID} || S_D || S_{HS})$. Then, the *HS* sends N_{HS} and S_{HS} to the device D_i, wherein the device D_i computes $W' = W \oplus h(P_D)$ and $K_{LAID} = h(W' || (nonce_D * S_{HS}))$ and verifies the received N_{HS} with the computed value of $h(K_{LAID} || S_D || S_{HS})$. This verification will only hold true if the device D_i has calculated the correct values of W'. Hence, the

HS ensure that the device D_i has computed correct K_{LAID} and the device also ensure that the *HS* has computed correct K_{LAID}. Therefore, LAID algorithm preserves mutual authentication.

Proposition 12 *LAID algorithm preserves non-traceability.*

Proof In LAID algorithm, the device D_i computes $S_D = nonce_D * S$ and the *HS* computes $S_{HS} = nonce_{HS} * S$. Based on $nonce_D$ and $nonce_{HS}$, the device D_i and the *HS*, further, compute K_{LAID} respectively without sharing private key directly. Since, $nonce_D$ and $nonce_{HS}$ are not fixed values but pseudorandom numbers randomly generated by them respectively, hence, due to randomness, the LAID algorithm preserves non-traceability.

Proposition 13 *LAID algorithm is resistant to offline password guessing attack.*

Proof In offline dictionary attack, any adversary $\mathscr{A}$ do not require to communicate with the legitimate entities and do not have lack of computing resources. The messages transmitted between the *HS* and the device D_i do not help the adversary $\mathscr{A}$ to determine and validate the guessed password, due to the unavailability $nonce_{HS} * nonce_D * S$. Therefore, LAID algorithm prevents offline password guessing attack.

Proposition 14 *LAID algorithm is resistant to replay attack.*

Proof In replay attack, the adversary $\mathscr{A}$ can keep resending the messages between the device D_i and the *HS*. In LAID algorithm, nonces are used to prevent replay attack. The device D_i selects $nonce_D$ and computes $S_D = (nonce_D \times S)$ and then sends to the *HS*. The *HS* selects $nonce_{HS}$ and computes $S_{HS} = (nonce_{HS} \times S)$. Due to pseudorandom nonces, LAID algorithm is resistant to replay attack.

Proposition 15 *LAID algorithm is resistant to side channel attack.*

Proof J.S. Coron. mentioned that the side channel attack can be prevented by randomness of mathematical parameters of ECC point multiplication[66]. Since, in LAID algorithm, we have used ECC, hence, LAID algorithm is secure against side channel attacks.

Proposition 16 *LAID algorithm is resistant to smart device capture attack.*

Proof When the device D_i is lost or captured by any adversary $\mathscr{A}$, the adversary $\mathscr{A}$ do not have access to the device D_i without knowledge of the password P_D. The adversary $\mathscr{A}$ cannot change the password without knowledge of the password P_D. Thus, smart device capture attack is prevented in LAID algorithm.

Proposition 17 *LAID algorithm is resistant to stolen-verifier attack.*

Proof The algorithm doesn't store the user's password directly on the server and use of random values $nonce_{HS}$ and $nonce_D$ for each session ensure that even if an adversary $\mathscr{A}$ steals the verifier data, it can't directly use the data to authenticate without knowing session-specific values. Therefore, the LAID algorithm has resistance against the stolen-verifier attack.

Results and performance analysis

Functional analysis and comparison

In this section, we discuss and compare different existing algorithms/schemes with the proposed LAID algorithm. We present comparative analysis based on functionality and resistance to different attacks. Table 3 below states that the proposed algorithm provides more security as it can withstand the mentioned security attacks and preserves the different security functionalities. The symbol ✓ indicates that the algorithm/scheme can withstand the respective attack or the respective security functionality is present. The symbol × indicates that the algorithm/scheme cannot withstand the respective attack or the respective security functionality is either not present or not mentioned.

Cost analysis

In this subsection, we perform comparative analysis based on computational cost, communication cost and energy consumption. For comparative analysis, we perform the comparison with existing state-of-the-art schemes in different paradigms of smart cities as presented in systematic literature review done in this paper. The proposed LAID algorithm is compared with existing schemes in different paradigms of the smart cities. Out of the compared schemes, scheme of[14] do not employ ECC operations and hence, show less computational time due to non-ECC operations but resulting in lesser security strength at specific key length. Computational time of different mathematical and cryptographic functions employed in the schemes have been referred from[16,51,53] and shown in Table 4.

Based on Table 4 above, the computational time of LAID algorithm can be calculated. In LAID algorithm, initialization phase does not incur any computational cost; device setup phase incurs only 1 symmetric key cryptographic operation and 2 hash operations at the *HS*; authentication phase incurs 2 elliptic curve point multiplications and 3 hash operations at the device and 2 elliptic curve point multiplications, 3 hash and 1 symmetric key cryptographic operation at the *HS* respectively; password-change phase incurs only 2 hash operations at device. Consequently, total of all these computations for our LAID algorithm results into 4 elliptic

	14#	19	20	30#	31	38#	40#	41	43	44	46	47#	54#	LAID
Brute force attack	×	✓	✓	×	×	×	×	✓	×	×	×	✓	✓	✓
Cookie theft attack	×	✓	✓	×	✓	×	×	✓	✓	×	×	×	×	✓
Data confidentiality	✓	✓	✓	✓	×	✓	✓	✓	×	×	✓	×	×	✓
DoS attack	✓	×	×	×	×	×	✓	×	×	×	×	✓	✓	✓
Device anonymity	✓	✓	✓	×	×	×	✓	×	✓	×	✓	×	✓	✓
Forward and Backward Secrecy	✓	✓	✓	✓	×	✓	✓	×	×	×	×	✓	✓	✓
Impersonation Attack	×	×	×	✓	✓	✓	×	×	✓	×	✓	✓	✓	✓
Insider attack	×	×	✓	×	✓	×	✓	×	×	×	✓	×	✓	✓
Key freshness property	✓	×	×	×	✓	✓	✓	×	✓	✓	✓	×	✓	✓
MITM attack	×	✓	✓	✓	×	✓	✓	✓	✓	✓	×	✓	✓	✓
Mutual authentication	×	✓	✓	×	✓	×	✓	×	✓	✓	✓	×	✓	✓
Non traceability	×	×	×	×	×	×	×	×	✓	×	×	×	✓	✓
Offline password guessing attack	×	✓	✓	✓	✓	×	×	✓	×	×	×	✓	✓	✓
Parallel session attack	✓	×	×	×	✓	✓	✓	×	✓	✓	✓	×	✓	✓
Replay attack	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	×	✓
Side channel attack	×	×	×	×	×	×	×	×	×	×	×	×	×	✓
Smart device capture attack	✓	×	✓	✓	✓	×	✓	×	✓	×	×	✓	✓	✓
Stolen-verifier attack	✓	✓	✓	×	×	✓	×	×	✓	✓	×	×	×	✓

Table 3. Functionality comparison. # These schemes do not employ ECC operations.

Notation	Description	Time (in ms)
T_{ECM}	Elliptic curve point multiplication	0.294
T_{SYM}	Symmetric key cryptographic operation	0.021
T_{HASH}	One-way hash operation	0.003
T_{PUF}	PUF	0.353
T_{PRNG}	PRNG	0.0823
T_{EXP}	Exponential function	0.16
T_{HECMP}	Single hyper elliptic curve multiplication	0.48
T_{ESM}	Elliptic curve scalar multiplication [51]	2.26
T_{ED}	Elapsed time for execution [53]	0.13

Table 4. Computational time of mathematical and cryptographic functions[16,51,53].

Compute Operation	No. of compute operations	Time (in ms)
Elliptic curve point multiplication	4	$(4^* T_{ECM}) = 1.176$
Symmetric key cryptographic operation	2	$(2^* T_{SYM}) = 0.588$
Hash operation	10	$(10^* T_{HASH}) = 0.03$
Total		**1.248**

Table 5. Computational time calculation of LAID Algorithm.

curve point multiplications, 2 symmetric key cryptographic operations and 10 hash operations. Therefore, the computational time incurred in LAID algorithm is calculated and presented in Table 5 below.

Since, computational time for XOR and concatenation operation is negligible, hence, their time have not been considered while doing computational cost analysis. Comparison of computational cost of LAID algorithm with other schemes[14,44,46,51,53] are presented in Table 6. Computational costs of other schemes have been taken from cited papers respectively.

As shown[14], shows 1.090 ms computation time respectively, however, it does not employs ECC operations, hence, resulting in lesser security strength at specific key length. From above comparative analysis, it is evident that the proposed LAID algorithm shows remarkable improvement of 28.52% from the closest counterpart of 1.604 ms of scheme presented in[44].

Comparison of communication cost is performed based on the number of messages exchanged between the HS and the device. In authentication phase, for establishing K_{LAID}, LAID algorithm exchanges HSD_{MSG} of 128 bits, S_D of 326 bits, N_{HS} of 128 bits, S_{HS} for 326 bits, and N_D for 128 bits. Therefore, during authentication phase,

References	Computational Cost (in ms)
14	$(8*T_{HASH}) + (1*T_{PRNG}) + (2*T_{PUF}) = 1.090$
44	$(4*T_{ECM}) + (5*T_{HASH}) + (2*T_{SYM}) = 1.604$
46	$(6*T_{HECMP}) = 2.880$
51	$(23*T_{HASH}) + (4*T_{ESM}) = 8.956$
53	$(12*T_{HASH}) + (14*T_{ED}) = 1.89$
LAID	1.248

Table 6. Comparison of computational Cost.

References	Communication Cost (in bits)
14	1536
44	2408
46	1440
51	2496
53	2080
LAID	1036

Table 7. Comparison of communication Cost.

Reference	Energy Consumption (in μJ)
14	26.16
44	38.496
46	69.12
51	214.94
53	45.36
LAID	29.952

Table 8. Comparison of energy Consumption.

total communication cost is 1036 bits. Table 7 below shows comparison of LAID algorithm with other schemes[14,44,46,51,53]. Communication costs of other schemes have been taken from cited papers respectively.

From above comparative analysis, it is evident that the proposed LAID algorithm shows reduction of communication overhead by 132.43% from the closest counterpart of 2408 bits presented in[44]. Compared to[14,46,51,53]the proposed LAID algorithm reduces communication overhead by 48.26%, 38.99%, 140.92% and 100.77% respectively.

Further, efficiency and sustainability of LAID algorithm can be determined based on the energy consumption. It is obvious to state that sustainability of security algorithm depends on amount of energy consumed provided the security strength is not compromised. In LAID algorithm, energy consumption is calculated based on the expression:

$$E = V * i * t \text{ (Joules)}. \tag{1}$$

For IoT devices, I = 8 mA and V = 3.0 V as specified in[67] and detailed discussion is in[68] and t = computational time. Therefore, we calculated energy consumption of LAID algorithm as in Eq. (1), for computational time of LAID, t_{LAID}=1.248 ms, therefore,

$E = (3*8*10^{-3} * 1.248*10^{-3})$ Joules = $29.952 *10^{-6}$ Joules = 29.952 μJoules or 29.952 μJ.

Comparison of energy consumption based on Eq. (1) is presented in Table 8.

Notably, energy consumption of[14] is 26.16 μJ and has lesser energy consumption than LAID algorithm due to their lower computational time since the scheme does not employ ECC operations. Remarkably, energy consumption of LAID algorithm is 29.952 μJ employing ECC operations also.

Hence, the proposed LAID algorithm shows remarkable achievement of computational cost of 1.248 ms, communication cost of 1036 bits and energy consumption of 29.952 μJ. Table 9 below presents cost metrics of LAID algorithm collectively.

Figures 6, 7 and 8 below shows graphical representation of comparison between related schemes with LAID algorithm based on computational cost, communication cost and energy consumption respectively.

Cost Metrics	LAID Algorithm
Computational Cost	1.248 ms
Communication Cost	1036 bits
Energy Consumption	29.952 μJ

Table 9. Cost metrics of LAID Algorithm.

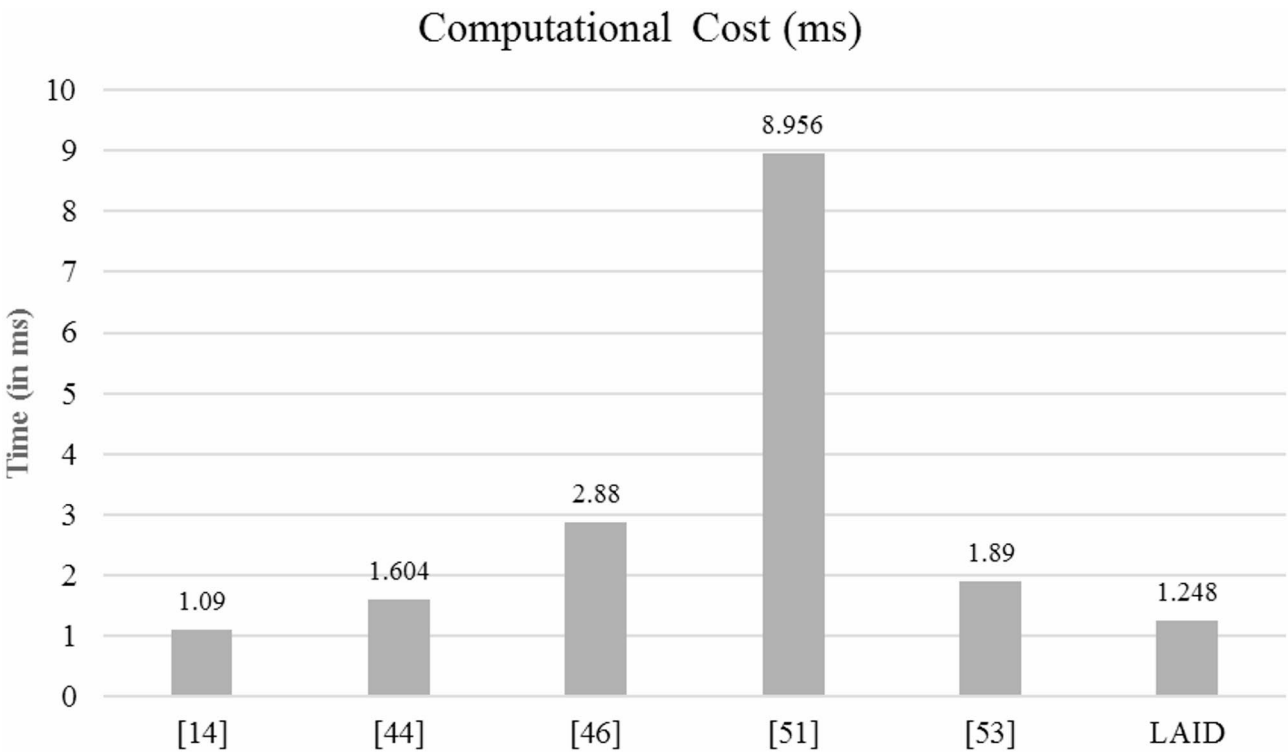


Fig. 6. Graphical summary of Computational Cost Comparison.

Conclusion and future work

Use of IoT devices is pervasive in sustainable smart cities. In this paper, we have identified need of authentication mechanisms among IoT devices in different paradigms of smart cities. Due to limited capabilities of IoT devices, it is challenging to provide secure authentication. Since, sustainable smart cities focus on renewable energy sources and its efficient usage, there is a need of devising approaches with less computational and communication overhead along with maintaining the balance of security and sustainability without compromising either of them. In this paper, we performed a detailed analysis of existing state-of-the-art using SLR approach and proposed a lightweight authentication mechanism for IoT devices (called as LAID) in SSC utilizing ECC. We perform automated formal analysis of validation using AVISPA simulation tool and informal security analysis to demonstrate that the proposed LAID algorithm is secure against different attacks. Although, our algorithm is resilient to several attacks as discussed earlier, however, reliance on secure pseudorandom number generation is a challenging limitation. We compared the proposed LAID algorithm with related work and performed the computational cost, communication cost and energy consumption analysis to assess the performance of our scheme. In cost analysis, the proposed LAID algorithm shows remarkable achievement of computational time of 1.248 ms, communication cost of 1036 bits and energy consumption of 29.952 μJ. Future work may include adaptive authentication systems that dynamically adjust security based on risk levels and privacy preservation.

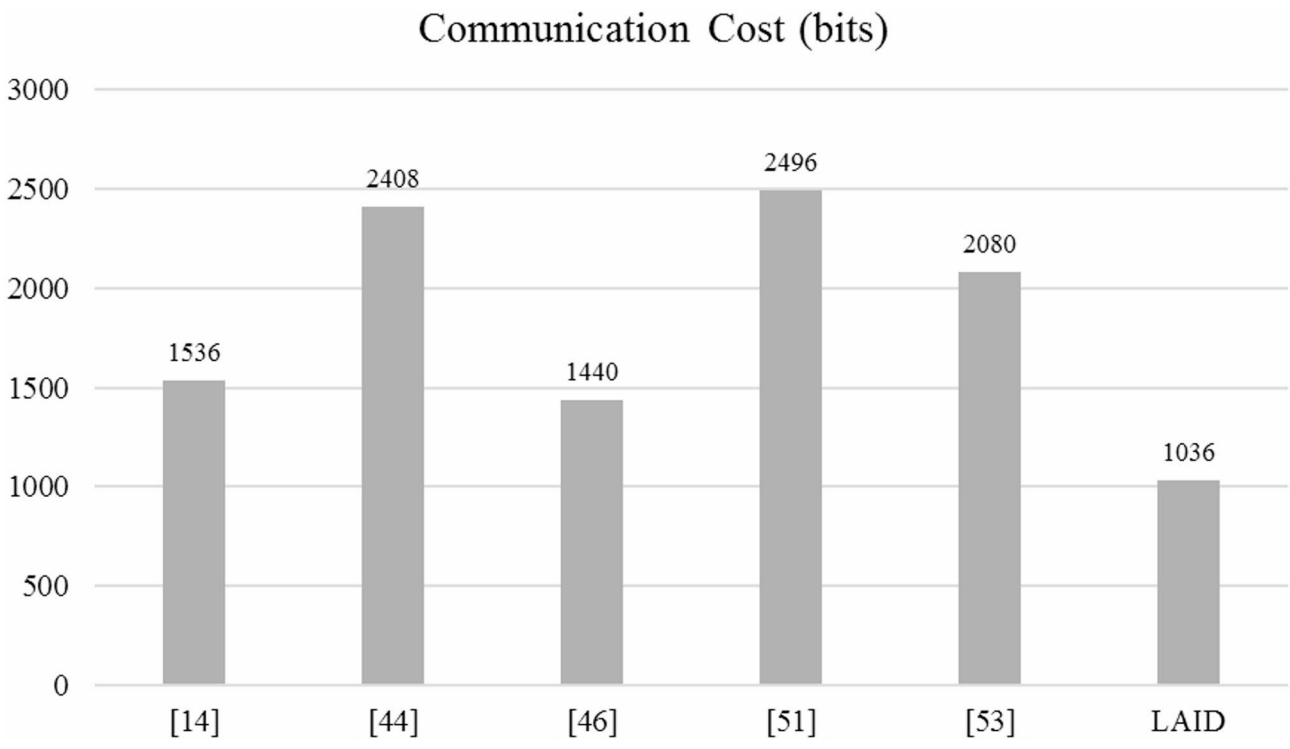


Fig. 7. Graphical summary of Communication Cost Comparison.

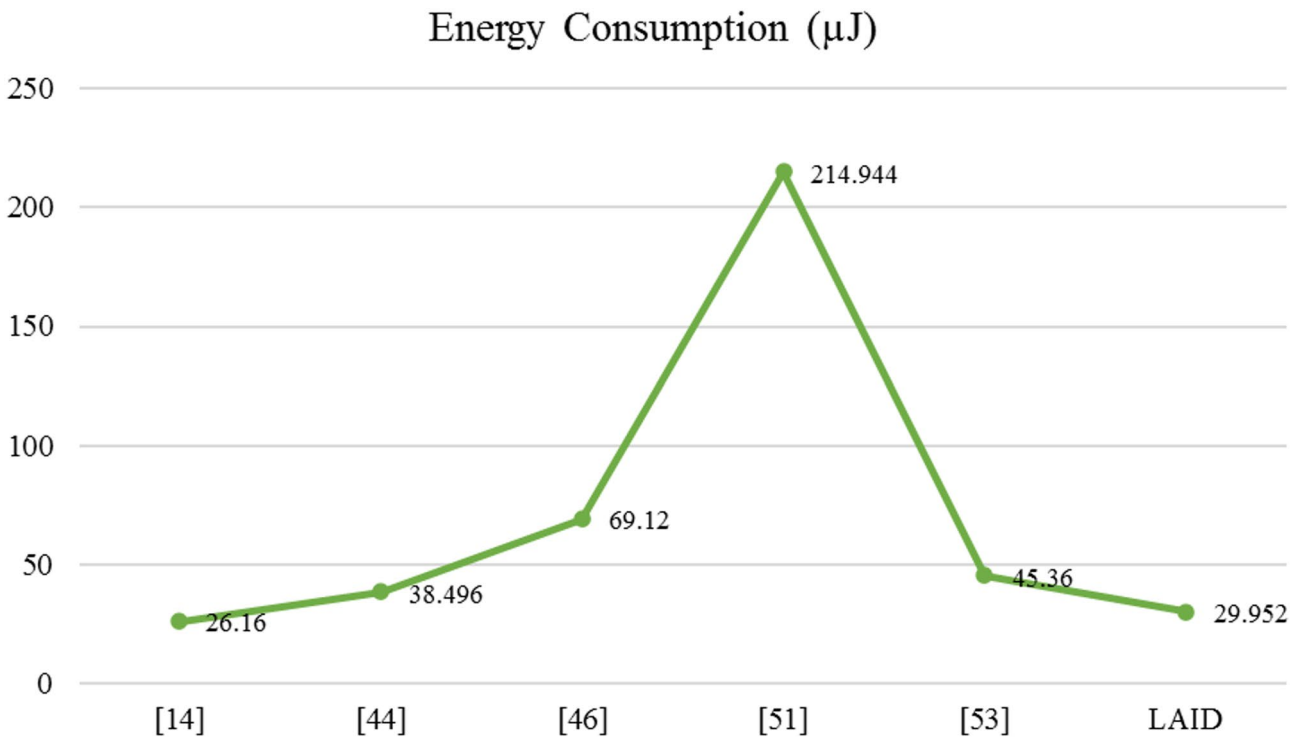


Fig. 8. Graphical summary of Energy Consumption Comparison.

Data availability

Data/Algorithm is provided within the manuscript and the complete code can be provided on request. Requests for algorithm/code should be addressed to Aqeel Khalique (Email: aqeelkhalique@gmail.com).

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Author contributions
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Declarations

Competing interests
The authors declare no competing interests.

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